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SHAPED CHARGE WITH REACTIVE MATERIAL

Abstract

Some implementations include an apparatus comprising an explosive material configured to detonate; a liner in contact with the explosive material, the liner configured to, after detonation of the explosive material, form jets to perforate a tubular disposed in a subsurface borehole; a reactive material configured to release energy in response to detonation of the explosive material; and a case in contact with a portion of the reactive material and configured to surround the explosive material, the liner, and the reactive material.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure generally relates to the field of subsurface operations and, more specifically,

to a shaped charged that may be used in subsurface hydraulic fracturing operations.

BACKGROUND

[0002] Shaped charges may be used in operations for hydraulic fracturing (“fracking”). Prior to hydraulic fracking operations, a wellbore may be drilled into a subsurface rock formation and lined with casing. Shaped charges may be used to create perforations in wellbore casing and the formation. These perforations may serve as pathways for hydraulic fracturing fluids to flow from the wellbore into the rock to create fractures in the rock.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the disclosure may be better understood by referencing the accompanying drawings.

[0004] FIG. 1 is a diagrammatic illustration of a shaped charge.

[0005] FIG. 2 is a diagram illustrating forces from detonating the shaped charge.

[0006] FIGS. 3-6 are diagrammatic illustrations of shaped charges.

[0007] FIG. 7 is an illustration depicting an example well system, according to some implementations.

[0008] FIG. 8 is a flow diagram illustrating operations for an example method.

DESCRIPTION

[0009] The description that follows includes example systems, methods, techniques, and operational flows that embody aspects of the disclosure. However, this disclosure may be practiced without these specific details. For clarity, some well-known structures and techniques have been omitted.

Overview

[0010] Shaped charges may be used for fracking and other operations for resource recovery. Traditionally, shaped charges may include an outer case surrounding an explosive material and a liner. The explosive material may be sandwiched between the case and the liner. As the explosive material detonates, there may be reactive forces created by explosive gases contacting the case. Some implementations increase these reactive forces by adding a layer of reactive material in contact with the case (such as by lining a portion of the case with reactive material). The reactive material may generate increased pressures in the wellbore and perforations in the formation. The reactive material may generate additional pressure near the case, thus increasing the amount of energy imparted on the liner--more than would result from merely adding mass to the case. As a result, the liner may collapse and jet at higher velocities than in traditional shaped charges. Some implementations may configure the geometry of the reactive material, explosive material, and other components to achieve various results. For example, placing reactive material near the apex of a conoidal case may create certain energy waves, pressures, and temperatures whereas placing the reactive material in other locations may achieve different energy waves, pressures, temperatures, and other conditions.

Description of Some Implementations

[0011] FIG. 1 is a diagrammatic illustration of a shaped charge. In FIG. 1, the shaped charge **100** may include a liner **108**, explosive material **106**, reactive material **104**, and a case **102**. The case **102** may be configured to surround the liner **108**, explosive material **106**, and reactive material **104**. The case **102** may be comprised of nonreactive material such as steel or other suitable materials. In some implementations, the case **102** itself may include a reactive material such as zinc.

[0012] The case **102** also may include an orifice **110** that may accommodate a detonator or other device used with the shaped charge **100**. In some implementations, the case **102** is conoidal but may take any suitable geometric form.

[0013] The liner **108** may be conoidal or any suitable geometric form. The liner **108** may be constructed from tungsten, copper, brass, aluminum, blended powder metals, or other suitable materials. The explosive material **106** may include one or more of the following explosive materials: Cyclotetramethylene-tetranitromine, Cyclotrimethylenetrinitramine, Hexanitrostilbene, pentaerythritol tetranitrate, Hexanitrohexaazaisowurtzitane, triaminotrinitrobenzene, 2,2',2'',4,4',4'',6,6',6''-Nonanitro-m-terphenyl, and 2,6-Bis (picrylamino)-3,5-dinitropyridine. The explosive material **106** may be conoidal or any suitable geometric form.

[0014] The reactive material **104** may be placed on an interior surface of the case **102**. In some implementations, the reactive material **104** may include one or more of the following materials: tin, chromium, aluminum, magnesium, titanium, tungsten, and tantalum. Other suitable reactive materials that are not metal include certain rubbers, polyethylene, polytetrafluoroethylene (e.g., TEFLON). The reactive material **104** may include any combination of these materials and any other suitable materials. The reactive material **104** be made from materials that are typically inert.

[0015] Operation of the shaped charge **100** may be initiated by detonating the explosive material **106**. As the explosive material **106** detonates, the explosion creates forces acting on the components of the shaped charge **100**. The following variables and equations represent aspects and forces that relate to detonation of the explosive material **106** in the shaped charge **100**. [0016] M—mass of the liner **108** [0017] C—mass of the explosive material **106** [0018] E—energy per mass of the explosive material **106** [0019] N—mass of the case **102** [0020] A—Gurney constant relating the velocity of the liner **108** to the detonation properties of the explosive material **106**

$$[00001] V_M = \sqrt{2E} \left[\frac{1+A^3}{3(1+A)} + \frac{N}{C} A^2 + \frac{M}{C} \right]^{-\frac{1}{2}} \quad (1) \quad V_N = A V_M \quad (2) \quad A = \frac{1 + \frac{2M}{C}}{1 + \frac{2N}{C}} \quad (3)$$

[0021] Equations 1-3 indicate that an increase in the mass of the case **102** (N) may cause a decrease to the Gurney constant (A) which may increase the velocity of the liner **108** (V.sub.M).

[0022] FIG. 2 is a diagram illustrating forces from detonating the shaped charge. As the explosive material **106** detonates, a reactive force **202** between the pressure of the explosive gases and the case **102** (N) may impart additional energy of the explosive gas on the liner (M). In some implementations, placing the reactive material **104** on an inner surface of the case **102** increases the reactive force by generating additional pressure near the case **102** (N), thus increasing the amount of energy imparted on the liner (M)—more than by merely adding additional mass to the case **102**. After detonation, the liner **108** may collapse and therefore jet at higher velocities than without the reactive material **104** placed in contact with the case **102**.

[0023] FIGS. 3-6 are diagrammatic illustrations of shaped charges. In each of FIGS. 3-6, the respective geometries of the reactive material **104** and explosive material **106** are different. The thickness, volume, and location of the reactive material **104** may vary depending on the desired results of the shaped charge **101**. In some instances, the geometric properties of the reactive material **104** may affect those of the explosive material **106**. In some embodiments (not shown in FIGS. 3-6), the geometry of any component may be modified to achieve desired results.

Irrespective of the geometries, placing a portion of the reactive material **104** in contact with the case may cause increase reactive forces inside the case **102** and achieve desired velocities, temperatures, and other effects. For example, a first configuration of the shape charge **100** with particular respective component geometries may be suitable to create reactive forces in the case **102** that create holes in a tubular (such as downhole casing) of a desired size. A second configuration with different respective component geometries may be suitable for deep holes in the formation. However, both configurations may include reactive material **104** in contact with the case **102**.

[0024] In FIG. 3, the reactive material **104** forms a layer on conoidal inner surface of the case **102**. The layer of reactive material **104** may span from an edge of the orifice **110** to a base of the conoidal inner surface. The layer of reactive material **104** may have uniform thickness.

[0025] In FIG. 4, the reactive material **104** spans from the orifice edge to the base of the conoidal

inner surface. However, in FIG. 4, the reactive material **104** does not have uniform thickness. In FIG. 4, the reactive material **104** is thickest near the orifice **110** while gradually thinning toward the base of the conoidal inner surface of the case **102**. In some implementations, the volume including the reactive material **104** and the explosive material is constant. For these implementations, as the amount of reactive material **104** increases, the amount of explosive material **106** decreases (and vice versa).

[0026] In FIG. 5, the reactive material **104** begins at the orifice edge and spans about $\frac{2}{3}$ the distance between the orifice **110** and the base of the conoidal inner surface of the case **102**. In FIG. 6, the reactive material **104** begins at the base of the conical inner surface of the case **102** and runs toward the orifice **110**. The reactive material **104** has tapered thickness at its ends with uniform thickness everywhere else. As noted, the geometries of the components in the shaped charge **100** may vary depending on the desired functionality. In some implementations, the reactive material **104** may take any suitable shape whereby at least a portion of the reactive material **104** is in contact with the case.

[0027] In some implementations, one or more shaped charges **100** may be utilized in various downhole scenarios. For example, one or more shaped charges **100** may be utilized for fracking in a well system. FIG. 7 is an illustration depicting an example well system, according to some implementations. In particular, FIG. 7 is a schematic of a well system **700** that includes a wellbore **702** in a subsurface formation **701**. The wellbore **702** may include a casing **706**. Shaped charges **100** may be used to create perforations **790A-790H** in the casing **706** at different depths. Although not shown, the shaped charges **100** may be utilized in concert with a perforation gun or other device for creating the perforations in the casing **706**. During fracking of the wellbores **702**, fracturing fluid (with or without sand) may be pumped into the subsurface formation **701** via the perforations **790A-790H**. The fracturing fluid may create fractures **750A-750H** in the subsurface formation such that reservoir fluid may flow into the wellbore **702**. In some implementations, the wellbore **702** may be hydraulically fractured in stages. For example, a first stage may include hydraulically fracturing the perforations **790G**, **790H** to generate fractures **750G**, **750H**, respectively. After fracking the first stage, a frac plug **730** may be positioned in the casing **706** above the first stage (i.e., at a lesser depth in the wellbore than perforations **790G**, **790H**). After setting the frac plug **730**, a sealing component (such as a ball, dart, etc.) may be positioned in the frac plug **730** to prevent hydraulic communication between the portion of the wellbore **702** below the frac plug **730** (i.e., the first stage that was hydraulically fractured) and the portion of the wellbore **702** above the frac plug **730**. Once the sealing component is positioned in the frac plug **730**, the perforations **790E**, **790F** may be formed in the casing **706** and hydraulic fracturing operations may commence for the next stage. Similar operations may be repeated for each subsequent stage (i.e., setting frac plug **732** and frac plug **734** and using the shaped charges **100** for hydraulically fracturing the next stage) until hydraulic fracturing operations for the wellbore **702** are complete.

[0028] FIG. 8 is a flow diagram illustrating operations for an example method. At block **802**, a shaped charge is inserted into a tubular inside a borehole, the shaped charge including a reactive material configured to release energy in response to detonation of an explosive material, wherein the reactive material is in contact with a case that surrounds the explosive material and the reactive material. At block **804**, the shaped charge is detonated to perforate the tubular.

[0029] As used herein, the term “or” is inclusive unless otherwise explicitly noted. Thus, the phrase “at least one of A, B, or C” is satisfied by any element from the set {A, B, C} or any combination thereof, including multiples of any element.

[0030] Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein but are to be accorded

the widest scope consistent with this disclosure, and the principles and the novel features disclosed herein.

[0031] The various implementations may include some implementations that have all or any combination of the aspects described herein.

[0032] Some implementations may aspects as described in the following clauses. [0033] Clause 1: An apparatus comprising: an explosive material configured to detonate; a liner in contact with the explosive material, the liner configured to, after detonation of the explosive material, form jets to perforate a tubular disposed in a subsurface borehole; a reactive material configured to release energy in response to detonation of the explosive material; and a case in contact with a portion of the reactive material and configured to surround the explosive material, the liner, and the reactive material. [0034] Clause 2: The apparatus of clause 1, wherein the reactive material includes at least one material selected from the group consisting of rubber, polyethylene, and

polytetrafluoroethylene. [0035] Clause 3: The apparatus of any one or more of clauses 1-2, wherein the reactive material is in further contact with a portion of the explosive material. [0036] Clause 4: The apparatus of any one or more of clauses 1-3, wherein the reactive material forms a layer in contact with an inner surface of the case. [0037] Clause 5: The apparatus of any one or more of clauses 1-4, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape. [0038] Clause 6: The apparatus of any one or more of clauses 1-5, wherein thickness of the reactive material increases from the orifice to the base. [0039] Clause 7: The apparatus of any one or more of clauses 1-6, wherein the reactive material is tapered at one or more edges. [0040] Clause 8: A system comprising: a shaped charge including an explosive material configured to detonate, a liner in contact with the explosive material, the liner configured to, after detonation of the explosive material, form jets to perforate a tubular disposed in a subsurface borehole, a reactive material configured to release energy in response to detonation of the explosive material, and a case in contact with a portion of the reactive material and configured to surround the explosive material, the liner, and the reactive material; and a perforation device configured to operate with the shaped charge to perforate the tubular in the subsurface borehole. [0041] Clause 9: The system of clause 8, wherein the reactive material includes at least one material selected from the group consisting of rubber, polyethylene, and polytetrafluoroethylene.

[0042] Clause 10: The system of any one or more of clauses 8-9, wherein the reactive material is in further contact with a portion of the explosive material.

[0043] Clause 11: The system of any one or more of clauses 8-10, wherein the reactive material forms a layer in contact with an inner surface of the case.

[0044] Clause 12: The system of any one or more of clauses 8-11, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape

[0045] Clause 13: The system of any one or more of clauses 8-12, wherein thickness of the reactive material increases from the orifice to the base.

[0046] Clause 14: The system of any one or more of clauses 8-13, wherein the reactive material is tapered at one or more edges.

[0047] Clause 15: A method comprising: inserting a shaped charge into a tubular inside a borehole, the shaped charge including a reactive material configured to release energy in response to detonation of an explosive material, wherein the reactive material is in contact with a case that surrounds the explosive material and the reactive material; and detonating the explosive material to perforate the tubular.

[0048] Clause 16: The method of clause 15, wherein the reactive material is in further contact with a portion of the explosive material.

[0049] Clause 17: The method of any one or more of clauses 15-16, wherein the reactive material forms a layer in contact with an inner surface of the case.

[0050] Clause 18: The method of any one or more of clauses 15-17, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape.

[0051] Clause 19: The method of any one or more of clauses 15-18, wherein thickness of the reactive material increases from the orifice to the base.

[0052] Clause 20: The method of any one or more of clauses 15-19, wherein the reactive material is tapered at one or more edges.

Claims

1. An apparatus comprising: an explosive material configured to detonate; a liner in contact with the explosive material, the liner configured to, after detonation of the explosive material, form jets to perforate a tubular disposed in a subsurface borehole; a reactive material configured to release energy in response to detonation of the explosive material; and a case in contact with a portion of the reactive material and configured to surround the explosive material, the liner, and the reactive material.
2. The apparatus of claim 1, wherein the reactive material includes at least one material selected from the group consisting of rubber, polyethylene, and polytetrafluoroethylene.
3. The apparatus of claim 1, wherein the reactive material is in further contact with a portion of the explosive material.
4. The apparatus of claim 1, wherein the reactive material forms a layer in contact with an inner surface of the case.
5. The apparatus of claim 4, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape.
6. The apparatus of claim 5, wherein thickness of the reactive material increases from the orifice to the base.
7. The apparatus of claim 4, wherein the reactive material is tapered at one or more edges.
8. A system comprising: one or more shaped charges each including an explosive material configured to detonate, a liner in contact with the explosive material, the liner configured to, after detonation of the explosive material, form jets to perforate a tubular disposed in a subsurface borehole, a reactive material configured to release energy in response to detonation of the explosive material, and a case in contact with a portion of the reactive material and configured to surround the explosive material, the liner, and the reactive material; and a perforation device configured to operate with the shaped charge to perforate the tubular in the subsurface borehole.
9. The system of claim 8, wherein the reactive material includes at least one material selected from the group consisting of rubber, polyethylene, and polytetrafluoroethylene.
10. The system of claim 8, wherein the reactive material is in further contact with a portion of the explosive material.
11. The system of claim 8, wherein the reactive material forms a layer in contact with an inner surface of the case.
12. The system of claim 11, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape.
13. The system of claim 12, wherein thickness of the reactive material increases from the orifice to the base.
14. The system of claim 11, wherein the reactive material is tapered at one or more edges.
15. A method comprising: inserting one or more shaped charges into a tubular inside a borehole, each shaped charge including a reactive material configured to release energy in response to detonation of an explosive material, wherein the reactive material is in contact with a case that surrounds the explosive material and the reactive material; and detonating the explosive material to perforate the tubular.

- 16.** The method of claim 15, wherein the reactive material is in further contact with a portion of the explosive material.
- 17.** The method of claim 15, wherein the reactive material forms a layer in contact with an inner surface of the case.
- 18.** The system of claim 17, wherein the case has a conoidal shape, and wherein the reactive material spans from an orifice at an apex of the conoidal shape to a base of the conoidal shape.
- 19.** The system of claim 18, wherein thickness of the reactive material increases from the orifice to the base.
- 20.** The system of claim 18, wherein the reactive material is tapered at one or more edges.
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